Analyzing IMDb Movie Scores with Machine Learning

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Abstract

Our project investigates if IMDb movie scores can be predicted or explained using movie metadata and financial features. Using a cleaned Kaggle dataset, we evaluated regression, classification, and clustering approaches. To predict exact ratings, we compared standard baselines against an improved linear regression model utilizing engineered financial features, release year, English-language status, and genre. We then reframed the problem to classify scores into low, medium, and high groups using a logistic regression model, alongside K-Means clustering for exploratory analysis. While our models outperformed their baselines, overall performance remained moderate. This suggests that movie scores are heavily influenced by subjective factors beyond the available metadata.

1 Introduction

The film industry continues to thrive due to the use of large-scale datasets that can analyze how the audience perceives media content. IMDb(1), a popular entertainment platform, provides numerical ratings for millions of movies and TV shows based on public opinion. This site allows reviews to be examined in an easy manner and have them influence people's viewing decisions. In addition to IMDb, movie studios release information about film production such as financial information and metadata associated with public opinion. Examining the relationship between these factors can provide clarity on how they can affect audience ratings.

This project focuses on the following question: To what extent can movie metadata and financial features explain or predict IMDb movie scores? Our goal is to determine what variables are strongly associated with higher audience ratings. Analyzing factors such as score, date, genre, and profit can help us view patterns with the data set and identify significant relationships. This project aims to utilize machine learning and statistical techniques to assess how these variables can predict IMDb scores. By using this way, filmmakers can find out what type of movies the audience enjoys and apply that information to future investments. The results of our project can help identify how a movie becomes successful and what measures can be taken to achieve that.

2 Related Work

Prior research in this field has utilized various machine learning techniques to predict box office performance using a specific structuralized movie metadata, audience reception of said movie, and overall movie success. Many of these studies focused on important categorical variables such as budget, genre, released date, runtime, popularity of cast members, and production company

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information in order to identify patterns that could be associated with successful films and high ratings from audiences.

2.1 Predictive Learning Modeling

Machine learning models such as XGBoost, XGBoost with Bagging(Bootstrap Aggregating), Random Forest, and Random Forest with Bagging were utilized by Kristianto et al. to explore movie success prediction(2). Their study's focus was on predicting movie success based on metadata collected and processed from IMDb datasets. XGBoost with Bagging achieved the highest performance with an 84.68% accuracy rating when compared to their other evaluation methods, demonstrating the effectiveness of combining various models to improve prediction performance. In addition to accuracy, the authors evaluated metrics such as precision, recall, F1 score, and Kappa coefficient that increase after tuning. The study demonstrated that movies can be grouped and classified by success level rather than their predicted exact ratings. Similarly, our project reframed IMDb score prediction as a classification problem by grouping movies into low, medium, and high score categories using quantile binning.

Various researchers have sought to and examined how, and if certain financial variables and movie metadata can in fact influence audience ratings. Chen and Antunovic analysed many variables, and found that movie budgets surprisingly showed only a weak negative correlation with IMDb ratings(3). This suggests that larger financial investments do not necessarily result in higher audience rating/scores. They also observed a similar trend when using imputed budget values, thus reinforcing the importance of careful evaluations of financial features during preprocessing.

2.2 Clustering Models

Prior studies have also applied the machine learning model K- Means to identify latent structure within movie datasets. These approaches have allowed the researcher to uncover and segment films based on similarity of financial performance, audience ratings and various genres without having the need of relying on labeled/observable outcomes. Such clustering models are able to find unseen patterns in movie success and inspect the grouping of high and low performing movies.

3 Methodology

3.1 Data Cleaning and Preprocessing

Our project uses the IMDb Movies Dataset from Kaggle(4), which contains information about movies, such as titles, release dates, user ratings, genres, language, budget, and revenue (5). We used `score` as the main target variable and built regression models to estimate the numerical score, classification models to predict score groups, and K-Means/PCA for exploratory clustering.

Before modeling, we cleaned and prepared the dataset. We renamed several columns to make them easier to work with, such as changing `names` to `title`, `date_x` to `release_date`, `budget_x` to `budget`, and `orig_lang` to `language`. We converted the release date column into a datetime format and then extracted the release year as a new feature. We also standardized categorical values such as genre, language, country, and status by replacing missing or empty values with `"Unknown"`. Only movies marked as released were kept, and rows with invalid scores or missing financial information were removed.

While exploring the data, we found that the `country` field did not appear to consistently represent a movie's actual production country. Many well-known non-Australian or international movies were labeled as `AU`, making the feature difficult to interpret. To avoid introducing misleading metadata into the models, we excluded `country` from the final regression, classification, and clustering feature sets.

For the supervised models, we split the cleaned dataset into 80% training data and 20% testing data. Preprocessing steps such as imputation and scaling were included inside scikit-learn pipelines so that they were fit on the training data and then applied to the test data.

3.2 Feature Engineering

We also created several new features from the original budget and revenue columns. Profit and return on investment (ROI) were defined:

$$\text{profit} = \text{revenue} - \text{budget}$$

$$\text{ROI} = \frac{\text{profit}}{\text{budget}}$$

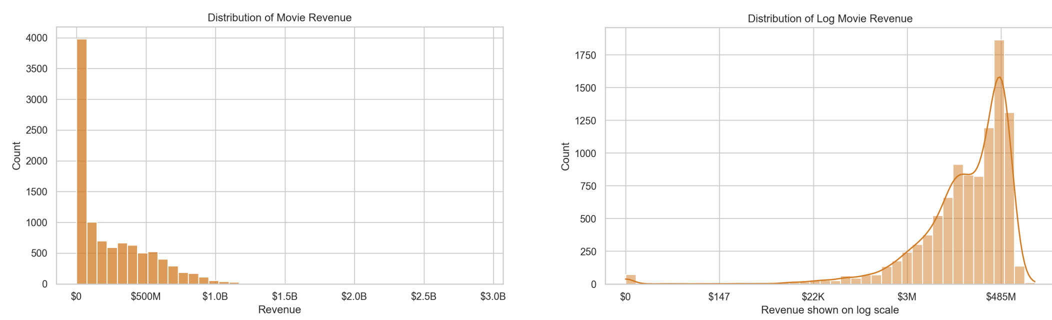
Since return on investment (ROI) can blow up when budget is very small, we limited the most extreme ROI values. We capped ROI at the 1st and 99th percentiles so unusually low or high values would not have too much influence on the models. Budget, revenue, and profit were also transformed because financial variables were highly skewed, as shown in Figure 1. A small number of blockbuster films can have extremely large budgets or revenues, so log transformations help make these variables more usable for linear models:

$$\log_{\text{budget}} = \log(1 + \text{budget})$$

$$\log_{\text{revenue}} = \log(1 + \text{revenue})$$

Since profit can be negative, we shifted profit before applying the log transform:

$$\log_{\text{profit shifted}} = \log\left(1 + (\text{profit} - \min(\text{profit}))\right)$$



(a) Raw revenue distribution

(b) Log revenue distribution

Figure 1: Comparison of raw revenue and log-transformed revenue distributions. The raw revenue feature is heavily right-skewed, while the log transformation reduces the effect of extreme blockbuster values.

We also created features from the language and genre columns. For language, we did not use every individual language as its own category. Instead, we created a binary feature called `is_english`, where English-language movies were coded as 1 and all other movies were coded as 0. This gave the model a simple way to capture whether a movie was in English without adding a large number of sparse language columns. This also matched the structure of the dataset, where English-language movies made up a large portion of the high-budget and high-revenue films.

For genre, we used multi-hot encoding because many movies have more than one genre. For example, a movie that is both action and adventure would receive a 1 in both the action and adventure genre columns, and a 0 in the others.

3.3 Regression Models

For the regression task, we compared three models. The first was a mean-score baseline, which simply predicts the average score from the training set for every movie. This gives a basic reference point for how well the other models are performing. The second model was a baseline linear regression model using only budget and revenue. The third model was an improved linear regression model using engineered financial features, including log-transformed budget and revenue, capped ROI, shifted log-profit, release year, `is_english`, and the genre columns. We did not include the detailed language column or the country column in the final model. The full language column would have added many sparse categories, and the country column appeared unreliable. We evaluated the regression models using root mean squared error (RMSE), mean absolute error (MAE), and R^2. RMSE and MAE measure average prediction error, while R^2 measures how much variation in movie scores is explained by the model.

3.4 Classification Models

We also approached the data with classification. Instead of trying to predict the exact score, we divided movies into three score categories using quantile binning: low, medium, and high. We compared a majority-class baseline, which always predicts the most common score category, against a logistic regression classifier. The classification models were evaluated using accuracy, balanced accuracy, precision, recall, and F1-score. We also used a confusion matrix to see which score categories the model confused most often.

3.5 Clustering and PCA

Finally, we used K-Means clustering as an exploratory part of the project. The goal was not to predict IMDb score directly. Instead, we wanted to see whether movies formed broad groups based on financial scale, profitability, release year, and genre composition. Since K-Means is based on distances between points, the numerical features were standardized before clustering.

We compared candidate values of k from 2 to 10 using the elbow plot to measure cluster compactness and silhouette scores to measure cluster separation. We selected $k = 4$ for the final cluster summaries because it provided a practical balance between detail and interpretability. After clustering, we summarized each cluster using average score, median budget, median revenue, median profit, median ROI, median release year, English-language rate, and most common main genre. We then used PCA to project the processed feature space into two dimensions so the clusters could be visualized.

4 Experiments & Results

4.1 Predicting IMDb Scores

Table 1: Regression model performance

Model	RMSE	MAE	R^2
Mean score baseline	10.329	7.718	-0.001
Budget/revenue baseline	10.122	7.548	0.039
Improved linear regression	9.307	6.763	0.188

The improved linear regression model performed best among the main regression models. It reduced RMSE from 10.329 for the mean baseline and 10.122 for the budget/revenue baseline to 9.307. Its R^2 value also increased to 0.188, which suggests that the engineered financial features, release year, language indicator, and genre indicators helped explain more variation in IMDb score than the simpler baselines.

Regression coefficient interpretation. To better understand the improved linear regression model, we looked closely at its feature weights, or coefficients. Since the numerical features were standardized first, the coefficients give us a general sense of which features had stronger relationships with the model's predicted score. This comparison is not perfect because the model includes both standardized numeric variables and binary genre indicators, but it still helps show what the model was relying on. The model only captures simple linear trends, and some financial features, such as budget, revenue, and profit, are closely related to each other. Because of that, the coefficients can sometimes blur exactly which feature is doing the most work.

Table 2: Largest interpretable coefficients from the improved linear regression model

Feature	Coefficient
Animation genre	2.308
Drama genre	1.908
Log profit	1.702
Log budget	-1.469
Horror genre	-1.210
Release year	-0.726
Documentary genre	0.682
History genre	0.644
Music genre	0.628
Crime genre	0.572

Table 2 shows several reasonable patterns in the model. Log profit had a positive coefficient, while log budget had a negative coefficient, suggesting that more profitable movies were associated with higher predicted scores, but larger budgets alone did not necessarily lead to higher scores. Genre also appeared to matter. Animation and drama had some of the largest positive coefficients, while horror had one of the largest negative coefficients. This may reflect differences in how audiences rate certain types of movies, especially since genres like horror can be more polarizing. Release year also had a negative coefficient, which suggests that older movies tended to receive slightly higher predicted scores in this model. One possible explanation is that older movies remaining in the dataset may include more well-known or highly regarded films, but this should not be interpreted as a direct causal effect.

While experimenting, we also tested Ridge regression as a regularized version of the improved linear regression model. The improved linear regression model achieved an RMSE of 9.307 and an R^2 of 0.188, while Ridge regression produced nearly identical results. The GridSearchCV version selected a larger regularization strength, but its test RMSE slightly increased to 9.312. Because regularization did not improve the final test performance, we retained our improved linear regression model as the main regression model.

4.2 Classifying Score Categories

Table 3: Distribution of quantile-based score categories

Score category	Proportion
Low	0.368
Medium	0.302
High	0.331

For the classification task, we used quantile binning to create low, medium, and high score categories. This choice was made because fixed score cutoffs produced a less balanced split, with the medium-score group becoming too dominant. Quantile binning made the classification task more balanced and gave the majority-class baseline a fairer comparison point as shown in Table 3.

Table 4: Classification model performance for low, medium, and high IMDb score categories

Model	Accuracy	Balanced Acc.	Precision	Recall	F1
Majority-class baseline	0.368	0.333	0.123	0.333	0.179
Logistic regression	0.503	0.492	0.481	0.492	0.481

The logistic regression classifier performed better than the majority-class baseline, showing that the engineered features contain information that helps separate low, medium, and high scoring movies. Table 4 shows that accuracy increased from 0.368 to 0.503, and macro F1-score increased from 0.179 to 0.481. Precision, recall, and F1-score are reported using macro averaging.

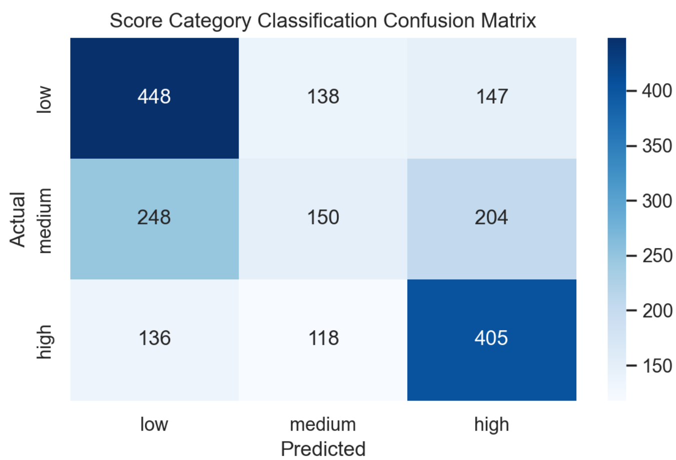


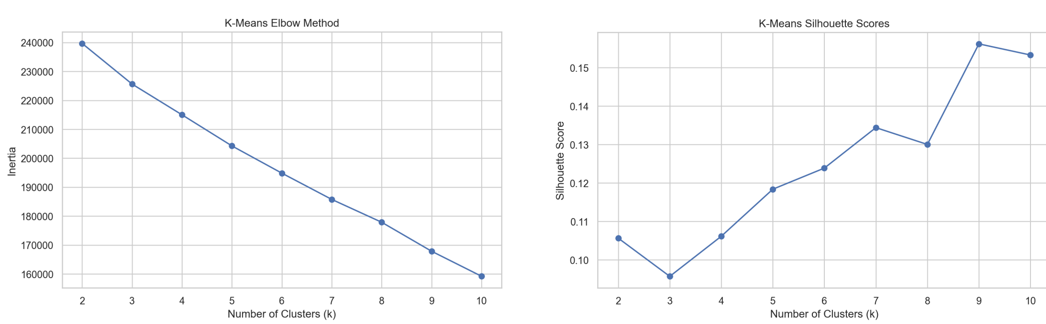
Figure 2: Confusion matrix for the logistic regression classifier on low, medium, and high IMDb score categories.

Figure 2 shows that the classifier was better at identifying movies in the low and high score groups than movies in the medium group. This makes sense because the medium category is less distinct: movies near the middle of the rating distribution can look similar to both lower-rated and higher-rated movies based on metadata and financial features alone. This suggests that the model captures some broad rating patterns, but exact score category boundaries remain noisy.

4.3 Exploring Movie Clusters

Table 5: K-Means cluster profiles using $k = 4$

Cluster	Count	Avg. Score	Median Budget	Median Revenue	Top Genre
0	2310	66.95	\$88.45M	\$329.86M	Animation
1	3575	62.74	\$27.00M	\$61.42M	Action
2	279	69.81	\$38.00M	\$81.60M	Drama
3	3803	65.17	\$51.27M	\$180.59M	Drama



(a) Elbow method

(b) Silhouette scores

Figure 3: Elbow and silhouette analyses for selecting the number of K-Means clusters.

Table 5 shows that the clusters mainly differed by financial scale and genre composition, with Cluster 0 having the highest median budget and revenue and Cluster 1 having the lowest average score. In Figure 3a, the elbow plot did not show a sharp bend, suggesting that there was no obvious best value of k. Figure 3b shows the silhouette scores were also fairly low overall, meaning the clusters were not strongly separated. Although larger values of k produced slightly higher silhouette scores, they did not make the cluster profiles much easier to interpret. For this reason, we selected $k = 4$ as a practical balance between detail and interpretability. The resulting clusters showed broad differences in financial scale, genre composition, and average score, but the PCA visualization also showed overlap between clusters. This suggests that K-Means was mostly useful as an exploratory tool rather than evidence of strongly separated movie categories.

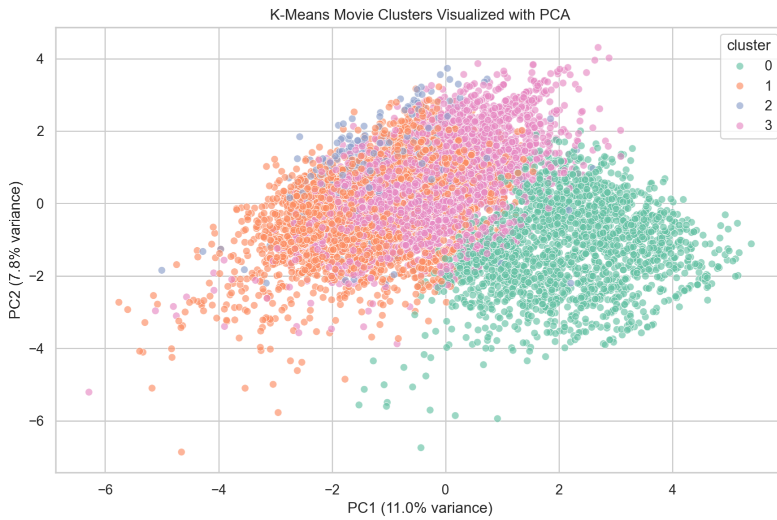


Figure 4: PCA visualization of K-Means movie clusters.

The first two principal components explain 18.8% of the total variance, so Figure 4 should be treated as a simplified visualization of the full clustering space.

5 Conclusion

Overall, this experiment exhibited the use of movie metadata and financial features to discover meaningful predictive signal about movie ratings. However, this relationship is moderately strong, revealing that the signal can be limited by subjective factors. Based on the regression approaches used, we found that the improved linear regression model performed the best. The model's RMSE decreased from the baselines to 9.307 while R^2 increased to 0.188. This observation shows that engineered features such as release year, genre indicators, language, and profitability were able to explain movie rating variations more thoroughly than simpler models. In addition to regression method, coefficient analysis displayed important trends with the data. First, more profitable movies tended to have higher predicted scores, displaying a direct, positive relationship between these two variables. Second, not all movies with larger budgets had better ratings, demonstrating how these variables are not associated with each other. In addition, genre impacted the results since drama and animation had higher scores while horror had lower scores. From these trends, we concluded that audience preference can have an impact on IMDb ratings. However, the R^2 value reveals a substantial amount of variation that cannot be explained by the variables tested. This means that external factors such as cast performance, screenplay, and marketing strength had an influence on results.

The results of the classification task verified the use of engineered features with the data. Logistic regression appeared to outperform the baseline of the majority-class, showing that this method was able to differentiate between low-, medium- and high-rated movies. In addition, the confusion matrix revealed some difficulty in classifying medium-rated movies, demonstrating an ambiguous area in the ratings. The K-Means clustering analysis highlighted that movies can be grouped by genre composition and financial scale instead of well-defined categories. However, the overlap in PCA visualization and low silhouette scores demonstrate that movies are not always separated into well-established clusters in the feature space. Therefore, we concluded that clustering is a useful exploratory tool for exploring the data instead of discovering movie types. The results obtained from this experiment reiterate the significant trends in movie ratings based on movie metadata and financial features. IMDb scores can be influenced by many factors, including those not included in the data set.

Future work. Future work could include more audience- and content-based features, such as cast popularity, director history, critic reviews, plot keywords, or sentiment from movie overviews and user reviews. It could also be useful to test more flexible models, such as Random Forests or gradient boosting, since these may capture nonlinear patterns that were not captured by the linear models used in this project. Another useful direction would be to compare audience-focused measures, such as IMDb scores, with aggregate critic scores. This could help show whether the same movie features are associated with both audience reception and critical reception, or whether those two types of reception follow different patterns. Future work could also examine box-office success more directly, such as by treating revenue, profit, or ROI as prediction targets instead of only using them as explanatory features. Together, these additions would give a fuller picture of movie reception, since a single numerical IMDb score may not fully capture how a movie is received.

Code Availability

The code and project files are available at:

`https://github.com/craighymo/imdb-movie-score-analysis`.

Note. All Code was initially generated via prompts provided to OpenAI Codex. ChatGPT and Google Gemini were also utilized to make further refinements. No code was handwritten.

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